Supplementary Material

Causal-Temporal Representation Learning for Comorbidity Progression Risk in Longitudinal Primary-Care Data

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S1. Data Processing and ICD-9 Macro-Category Mapping

Raw ICD-9-CM diagnosis codes were aggregated into a compact set of macro-categories following standard chapter boundaries. Supplementary and heterogeneous residual categories were grouped into SUPP, including V-codes, E-codes, pregnancy- and childbirth-related categories, ill-defined conditions, and residual non-standard diagnosis strings. The complete mapping used in the study is reported in Table 1.

Table 1. Mapping of ICD-9-CM diagnosis codes to CHRONOS macro-categories.

Macro	ICD-9-CM Range	Description
INFE	001–139	Infectious and Parasitic Diseases
NEOP	140–239	Neoplasms
ENDO	240–279	Endocrine, Nutritional and Metabolic Diseases
BLD	280–289	Diseases of the Blood and Blood-forming Organs
MENT	290–319	Mental Disorders
NERV	320–389	Diseases of the Nervous System and Sense Organs
CIRC	390–459	Diseases of the Circulatory System
RESP	460–519	Diseases of the Respiratory System
DIGE	520–579	Diseases of the Digestive System
GEN	580–629	Diseases of the Genitourinary System
SKIN	680–709	Diseases of the Skin and Subcutaneous Tissue
MUSC	710–739	Diseases of the Musculoskeletal System and Connective Tissue
CONG	740–759	Congenital Anomalies
PERI	760–779	Certain Conditions Originating in the Perinatal Period
INJ	800–999	Injury and Poisoning
SUPP	V/E, 630–679, 780–799	Supplementary and residual administrative categories

This aggregation serves two purposes. First, it reduces the dimensionality of the raw ICD-9-CM code space, which would otherwise be too sparse for large-scale edge-level target trial emulation. Second, it increases empirical support for candidate exposure–outcome pairs, making overlap diagnostics and weighted estimation more stable. Because `SUPP` is heterogeneous by construction, edges involving this category should be interpreted more cautiously than those involving clinically tighter groups such as `CIRC`, `DIGE`, or `GEN`.

S2. Cohort Summary and Descriptive Statistics

Table 2 provides an overview of the processed real-world primary-care cohort used throughout the study. Overall, the cohort includes 64,828 patients and more than 9 million longitudinal diagnosis events, indicating both substantial population coverage and dense event-level observation. The available records span a long calendar window, from 2000-01-02 to 2018-07-19, which supports the analysis of medium and long-term disease progression patterns.

The cohort is also characterized by within-patient longitudinal depth. Patients contribute a median of 42 diagnosis events, with a wide interquartile range (12–148), suggesting marked heterogeneity in healthcare utilization and clinical complexity. Median follow-up is 2,209 days (IQR 955–3,629), corresponding to approximately six years of observation, which is consistent with the longitudinal nature required for progression-oriented analyses.

From a demographic perspective, the median age at first available record is 44 years (IQR 28–60). Sex distribution is broadly balanced, with 34,298 female patients (52.9%) and 30,511 male patients (47.1%), while only 19 records carry residual non-standard sex labels. Finally , the most represented macro-categories by patient coverage are `SUPP`, `RESP`, `DIGE`, `MUSC`, and `GEN`, indicating that the cohort is dominated by broad chronic-care and frequently documented primary-care domains. Taken together, these descriptive statistics suggest that the processed cohort is sufficiently large, temporally rich, and clinically heterogeneous to support the downstream causal and longitudinal analyses.

Figure 1 reports the distribution of diagnosis events across ICD-9 macro-categories. The observed imbalance motivates both macro-category aggregation and edge-specific support diagnostics in the main causal analysis.

S3.Causal Identification Assumptions

Our formulation of the problem follows the Rubin–Neyman potential-outcomes framework [1,2], adapted here to edge-specific target trial emulation. Causal effects are defined through contrasts between these counterfactual outcomes and are identified from observational data under the following assumptions.

First, we assume the Stable Unit Treatment Value Assumption (SUTVA), which in our setting combines consistency and no interference. Consistency

Table 2. Descriptive statistics of the processed real-world primary-care cohort.

Characteristic	**Value**
Number of patients	64,828
Number of longitudinal diagnosis events	9,063,742
Calendar span of records	2000-01-02 – 2018-07-19
Median events per patient (IQR)	42 (12–148)
Median follow-up in days (IQR)	2,209 (955–3,629)
Median age at first available record, years (IQR)	44 (28–60)
Female, n (%)	34,298 (52.9%)
Male, n (%)	30,511 (47.1%)
Residual non-standard sex labels, n	19
Most represented macro-categories by patient coverage	SUPP, RESP, DIGE, MUSC, GEN

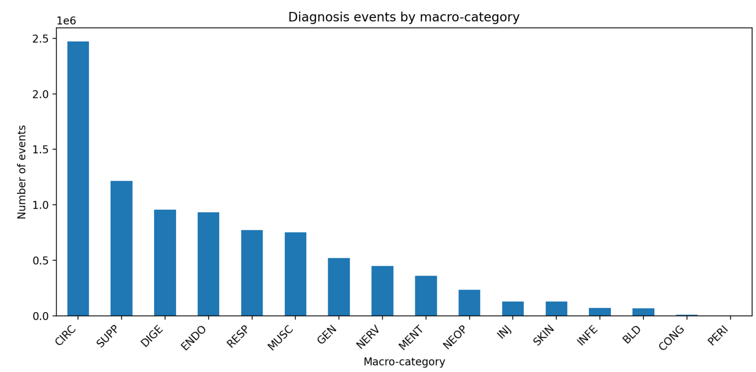

Fig. 1. Distribution of diagnosis events across ICD-9 macro-categories in the processed cohort. Event counts are markedly imbalanced across categories, with the highest frequencies observed in `CIRC`, `SUPP`, `DIGE`, `ENDO`, and `RESP`.

means that if patient i is observed under treatment status $T_i(t_0) = t$, then the observed outcome equals the corresponding realized potential outcome, i.e.,

$$T_i(t_0) = t \Rightarrow Y_i = Y_i(t), \qquad t \in \{0, 1\}.$$

This requires the event "incident onset of D_a at t_0" to be sufficiently well defined under the trial protocol, so that treatment corresponds to a consistent clinical event rather than to heterogeneous hidden versions of treatment. No interference means that one patient's treatment status does not affect another patient's outcome. In other words, whether one patient experiences the onset of D_a at time t_0 is assumed not to change the probability that another patient develops D_b within the follow-up window. In the present setting, this is a reasonable working assumption because the unit of analysis is the individual patient trajectory.

Second, we assume ignorability, or no unmeasured confounding, meaning that conditional on the measured pre-index history H_{t_0}, treatment assignment at time zero is independent of the potential outcomes:

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp T_i(t_0) \mid H_{t_0}.$$

Operationally, adjustment is performed through the edge-specific pre-index representation H_{t_0} used by the model, which combines measured static covariates, recent utilization, selected historical counts, and temporal event history available before t_0. This representation is used to estimate both the propensity score and the outcome regressions entering the cross-fitted AIPW estimator. Intuitively, after conditioning on this measured pre-index information, treated and control patients should be more comparable with respect to their counterfactual risk of developing D_b. However, this assumption cannot be verified directly, and residual confounding may remain because relevant factors such as latent clinical severity, physician judgment, or other unrecorded determinants may still affect treatment assignment and outcome risk.

We assume positivity, or overlap, meaning that for every clinically admissible pre-index history with positive density, both treatment states remain possible:

$$0 < \mathbb{P}(T_i(t_0) = 1 \mid H_{t_0}) < 1.$$

This means that, among patients with comparable observed histories, some may experience the incident onset of D_a at t_0 and some may not. In practice, edge-specific eligibility rules, risk-set sampling, and propensity trimming help improve the plausibility of this assumption, although they do not guarantee it.

S4. Complete $D_a \to D_b$ Edge Results

This section reports the complete screening results for all ordered $D_a \to D_b$ pairs evaluated under the edge-level target trial emulation framework. The table is intended primarily as a lookup resource: for each directed pair of ICD-9 macro-categories, it summarizes the estimated progression-risk relation after confounding adjustment, together with uncertainty, a residual-bias diagnostic, sample support, and the result of global multiple-testing correction.

Table 3 should be read row by row, with each row corresponding to one directed candidate relation from an exposure category D_a to an outcome category D_b. The `Exposure` and `Outcome` columns identify the source and target macro-categories, respectively. The `ATE` column reports the cross-fitted doubly robust risk-difference estimate for the effect of incident onset of the exposure category on the subsequent occurrence of the outcome category within the prespecified follow-up window. Positive values indicate that incident onset of D_a is associated with a higher downstream risk of observing D_b, after adjustment for measured confounding.

The `95% CI` column reports the corresponding Wald-type confidence interval and provides a first indication of estimation uncertainty. In general, rows with confidence intervals far from zero and relatively narrow width may be interpreted as more precisely estimated than rows with wider intervals. The `Placebo` column reports the estimated effect in the short post-index placebo window, which is used here as a residual-bias diagnostic rather than as a causal estimand of substantive interest. Values close to zero are more consistent with limited short-window residual bias, whereas larger placebo magnitudes suggest that the corresponding edge should be interpreted with greater caution even when the main ATE is statistically significant.

The column n denotes the total size of the edge-specific emulated sample after risk-set sampling, that is, the combined number of treated and control observations included in the corresponding target trial emulation. Accordingly, $n = n_T + n_C$, with treated and control counts shown in the n_T/n_C column. This quantity should not be interpreted as the number of unique patients in the original cohort. Because controls may be reused across treated anchors during risk-set sampling, and because the same individual may contribute to multiple eligible risk sets, the effective edge-specific sample size may exceed the number of distinct individuals in the source dataset.

Finally, the q-`value` column reports the Benjamini–Hochberg adjusted significance level across all tested $D_a \to D_b$ edges, while `FDR` indicates whether the edge is retained under the chosen false discovery rate threshold. For interpretation, the most convincing rows are typically those that combine four properties: a non-negligible positive ATE, a confidence interval that remains away from zero, a placebo estimate of limited magnitude, and a favorable adjusted significance level. An example is the row `BLD` $\to$ `CIRC`, for which the estimated ATE is 0.057 with 95% CI $[0.048, 0.066]$. This means that, in the emulated trial for this ordered pair, incident onset in the `BLD` category is associated with an estimated 5.7% point increase in the probability of subsequently observing `CIRC` within the analysis window. The confidence interval is relatively narrow and remains fully above zero, indicating a stable positive association under the adopted estimator. At the same time, however, the placebo estimate is -0.088, whose non-trivial magnitude suggests that this edge, although statistically strong, may still reflect some residual short-window structure and should therefore be interpreted more cautiously than edges with placebo values closer to zero.

More broadly, the full table is not meant to imply that every FDR-retained edge should be interpreted equally strongly. Rather, it provides a complete screening view from which readers can distinguish between edges that appear both statistically robust and diagnostically cleaner, and edges that are statistically detectable but potentially more sensitive to residual bias, sample construction, or temporal proximity effects.

Table 3: Complete $D_a \to D_b$ edge results.

Exposure	**Outcome**	**ATE**	**95% CI**	**Placebo**	n	n_T/n_C	q**-value**	**FDR**
BLD	CIRC	0.057	[0.048, 0.066]	-0.088	24,548	6,137/18,411	$< 10^{-12}$	Yes
BLD	CONG	0.005	[0.003, 0.007]	-0.000	32,536	8,134/24,402	1.45×10^{-6}	Yes
BLD	DIGE	0.084	[0.073, 0.095]	-0.029	27,300	6,825/20,475	$< 10^{-12}$	Yes
BLD	ENDO	0.119	[0.111, 0.128]	-0.035	28,944	7,236/21,708	$< 10^{-12}$	Yes
BLD	GEN	0.070	[0.058, 0.082]	-0.027	30,404	7,601/22,803	$< 10^{-12}$	Yes

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Exposure	**Outcome**	**ATE**	**95% CI**	**Placebo**	n	n_T/n_C	q**-value**	**FDR**
BLD	INFE	0.027	[0.021, 0.032]	-0.004	32,340	8,085/24,255	$< 10^{-12}$	Yes
BLD	INJ	0.022	[0.016, 0.028]	-0.002	32,076	8,019/24,057	2.19×10^{-12}	Yes
BLD	MENT	0.025	[0.018, 0.031]	-0.015	31,080	7,770/23,310	$< 10^{-12}$	Yes
BLD	MUSC	0.054	[0.044, 0.064]	-0.040	29,656	7,414/22,242	$< 10^{-12}$	Yes
BLD	NEOP	0.032	[0.026, 0.037]	-0.002	31,736	7,934/23,802	$< 10^{-12}$	Yes
BLD	NERV	0.029	[0.021, 0.038]	-0.019	31,172	7,793/23,379	2.79×10^{-11}	Yes
BLD	PERI	0.001	[0.000, 0.001]	-0.000	32,596	8,149/24,447	3.07×10^{-3}	Yes
BLD	RESP	0.054	[0.042, 0.066]	-0.039	30,064	7,516/22,548	$< 10^{-12}$	Yes
BLD	SKIN	0.027	[0.021, 0.033]	-0.005	32,112	8,028/24,084	$< 10^{-12}$	Yes
BLD	SUPP	0.129	[0.117, 0.140]	-0.030	27,876	6,969/20,907	$< 10^{-12}$	Yes
CIRC	BLD	0.004	[0.001, 0.007]	-0.004	73,079	18,270/54,809	9.84×10^{-3}	Yes
CIRC	CONG	0.001	[-0.001, 0.002]	-0.000	73,387	18,347/55,040	0.269	No
CIRC	DIGE	0.044	[0.035, 0.052]	-0.032	68,754	17,189/51,565	$< 10^{-12}$	Yes
CIRC	ENDO	0.037	[0.032, 0.042]	-0.032	69,627	17,407/52,220	$< 10^{-12}$	Yes
CIRC	GEN	0.028	[0.021, 0.035]	-0.014	71,339	17,835/53,504	$< 10^{-12}$	Yes
CIRC	INFE	0.005	[0.001, 0.009]	-0.005	73,099	18,275/54,824	0.012	Yes
CIRC	INJ	0.014	[0.010, 0.019]	-0.001	72,891	18,223/54,668	7.49×10^{-12}	Yes
CIRC	MENT	0.014	[0.010, 0.019]	-0.017	71,563	17,891/53,672	2.87×10^{-10}	Yes
CIRC	MUSC	0.004	[-0.004, 0.012]	-0.030	69,563	17,391/52,172	0.381	No
CIRC	NEOP	0.000	[-0.003, 0.004]	-0.007	72,631	18,158/54,473	0.855	No
CIRC	NERV	0.022	[0.015, 0.028]	-0.018	71,594	17,899/53,695	1.01×10^{-10}	Yes
CIRC	PERI	0.000	[0.000, 0.001]	-0.000	73,455	18,364/55,091	0.039	Yes
CIRC	RESP	0.043	[0.036, 0.050]	-0.022	70,102	17,526/52,576	$< 10^{-12}$	Yes
CIRC	SKIN	0.002	[-0.002, 0.006]	-0.004	72,895	18,224/54,671	0.292	No
CIRC	SUPP	0.072	[0.064, 0.080]	-0.025	67,347	16,837/50,510	$< 10^{-12}$	Yes
CONG	BLD	0.011	[-0.000, 0.023]	-0.002	5,828	1,457/4,371	0.060	No
CONG	CIRC	0.011	[-0.016, 0.038]	-0.115	4,640	1,160/3,480	0.437	No
CONG	DIGE	0.109	[0.082, 0.135]	-0.040	4,984	1,246/3,738	$< 10^{-12}$	Yes
CONG	ENDO	-0.015	[-0.037, 0.007]	-0.052	5,300	1,325/3,975	0.203	No
CONG	GEN	0.008	[-0.016, 0.031]	-0.021	5,528	1,382/4,146	0.538	No
CONG	INFE	-0.002	[-0.015, 0.011]	-0.004	5,920	1,480/4,440	0.780	No
CONG	INJ	-0.015	[-0.029, -0.001]	-0.003	5,864	1,466/4,398	0.049	Yes
CONG	MENT	-0.017	[-0.032, -0.002]	-0.017	5,732	1,433/4,299	0.033	Yes
CONG	MUSC	0.050	[0.023, 0.076]	-0.044	5,228	1,307/3,921	4.21×10^{-4}	Yes
CONG	NEOP	0.005	[-0.007, 0.017]	-0.002	5,844	1,461/4,383	0.430	No
CONG	NERV	0.041	[0.021, 0.060]	-0.024	5,636	1,409/4,227	5.38×10^{-5}	Yes
CONG	PERI	0.015	[0.013, 0.016]	0.000	5,960	1,490/4,470	$< 10^{-12}$	Yes
CONG	RESP	0.042	[0.016, 0.068]	-0.029	5,520	1,380/4,140	2.15×10^{-3}	Yes
CONG	SKIN	0.006	[-0.008, 0.020]	-0.003	5,876	1,469/4,407	0.437	No
CONG	SUPP	0.066	[0.039, 0.093]	-0.029	5,208	1,302/3,906	2.83×10^{-6}	Yes
DIGE	BLD	0.010	[0.007, 0.012]	-0.002	132,356	33,089/99,267	$< 10^{-12}$	Yes
DIGE	CIRC	0.022	[0.019, 0.026]	-0.091	117,902	29,476/88,426	$< 10^{-12}$	Yes
DIGE	CONG	0.001	[0.000, 0.002]	-0.000	132,820	33,205/99,615	0.031	Yes
DIGE	ENDO	0.007	[0.004, 0.011]	-0.041	126,660	31,665/94,995	8.28×10^{-6}	Yes
DIGE	GEN	0.019	[0.014, 0.023]	-0.018	129,952	32,488/97,464	$< 10^{-12}$	Yes
DIGE	INFE	0.010	[0.007, 0.013]	-0.004	132,448	33,112/99,336	1.47×10^{-10}	Yes
DIGE	INJ	0.013	[0.010, 0.016]	-0.003	131,992	32,998/98,994	$< 10^{-12}$	Yes
DIGE	MENT	0.015	[0.013, 0.018]	-0.016	130,572	32,643/97,929	$< 10^{-12}$	Yes
DIGE	MUSC	0.044	[0.039, 0.049]	-0.034	128,564	32,141/96,423	$< 10^{-12}$	Yes
DIGE	NEOP	0.011	[0.009, 0.013]	-0.003	131,972	32,993/98,979	$< 10^{-12}$	Yes
DIGE	NERV	0.029	[0.025, 0.033]	-0.016	130,287	32,572/97,715	$< 10^{-12}$	Yes
DIGE	PERI	-0.000	[-0.000, 0.000]	-0.000	132,896	33,224/99,672	0.550	No
DIGE	RESP	0.024	[0.019, 0.030]	-0.027	128,147	32,037/96,110	$< 10^{-12}$	Yes
DIGE	SKIN	0.008	[0.005, 0.011]	-0.004	132,196	33,049/99,147	2.04×10^{-6}	Yes
DIGE	SUPP	0.061	[0.054, 0.067]	-0.031	125,648	31,412/94,236	$< 10^{-12}$	Yes
ENDO	BLD	0.013	[0.010, 0.017]	-0.003	66,161	16,541/49,620	$< 10^{-12}$	Yes
ENDO	CIRC	0.086	[0.080, 0.091]	-0.100	53,037	13,260/39,777	$< 10^{-12}$	Yes
ENDO	CONG	0.000	[-0.001, 0.001]	-0.000	66,621	16,656/49,965	0.967	No
ENDO	DIGE	0.090	[0.082, 0.097]	-0.047	59,545	14,887/44,658	$< 10^{-12}$	Yes

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Exposure	**Outcome**	**ATE**	**95% CI**	**Placebo**	n	n_T/n_C	q**-value**	**FDR**
ENDO	GEN	0.033	[0.026, 0.039]	-0.021	63,853	15,964/47,889	$< 10^{-12}$	Yes
ENDO	INFE	0.008	[0.004, 0.011]	-0.003	66,273	16,569/49,704	9.64×10^{-5}	Yes
ENDO	INJ	0.006	[0.002, 0.010]	-0.004	66,021	16,506/49,515	4.06×10^{-3}	Yes
ENDO	MENT	0.028	[0.024, 0.032]	-0.017	64,345	16,087/48,258	$< 10^{-12}$	Yes
ENDO	MUSC	0.019	[0.012, 0.026]	-0.040	61,897	15,475/46,422	1.75×10^{-7}	Yes
ENDO	NEOP	0.010	[0.006, 0.014]	-0.003	65,785	16,447/49,338	1.62×10^{-7}	Yes
ENDO	NERV	0.008	[0.002, 0.013]	-0.020	64,413	16,104/48,309	6.74×10^{-3}	Yes
ENDO	PERI	0.000	[0.000, 0.001]	-0.000	66,697	16,675/50,022	0.022	Yes
ENDO	RESP	0.030	[0.022, 0.037]	-0.032	62,701	15,676/47,025	$< 10^{-12}$	Yes
ENDO	SKIN	0.012	[0.008, 0.016]	-0.005	65,973	16,494/49,479	1.74×10^{-8}	Yes
ENDO	SUPP	0.098	[0.090, 0.106]	-0.032	59,773	14,944/44,829	$< 10^{-12}$	Yes
GEN	BLD	0.012	[0.009, 0.015]	-0.002	111,876	27,969/83,907	$< 10^{-12}$	Yes
GEN	CIRC	0.025	[0.021, 0.029]	-0.111	95,544	23,886/71,658	$< 10^{-12}$	Yes
GEN	CONG	0.001	[-0.000, 0.002]	-0.001	112,400	28,100/84,300	0.314	No
GEN	DIGE	-0.004	[-0.009, 0.002]	-0.042	103,892	25,973/77,919	0.268	No
GEN	ENDO	0.060	[0.056, 0.064]	-0.049	105,476	26,369/79,107	$< 10^{-12}$	Yes
GEN	INFE	0.022	[0.019, 0.025]	-0.001	111,996	27,999/83,997	$< 10^{-12}$	Yes
GEN	INJ	0.017	[0.014, 0.021]	-0.004	111,800	27,950/83,850	$< 10^{-12}$	Yes
GEN	MENT	0.014	[0.011, 0.017]	-0.014	109,812	27,453/82,359	$< 10^{-12}$	Yes
GEN	MUSC	0.016	[0.010, 0.021]	-0.036	107,776	26,944/80,832	1.06×10^{-8}	Yes
GEN	NEOP	0.016	[0.013, 0.019]	-0.002	111,460	27,865/83,595	$< 10^{-12}$	Yes
GEN	NERV	0.031	[0.027, 0.035]	-0.018	109,672	27,418/82,254	$< 10^{-12}$	Yes
GEN	PERI	-0.000	[-0.000, 0.000]	-0.000	112,456	28,114/84,342	0.372	No
GEN	RESP	0.035	[0.029, 0.041]	-0.035	108,012	27,003/81,009	$< 10^{-12}$	Yes
GEN	SKIN	0.013	[0.009, 0.016]	-0.005	111,824	27,956/83,868	$< 10^{-12}$	Yes
GEN	SUPP	0.075	[0.068, 0.081]	-0.035	104,576	26,144/78,432	$< 10^{-12}$	Yes
INFE	BLD	0.010	[0.007, 0.014]	-0.003	57,012	14,253/42,759	8.90×10^{-9}	Yes
INFE	CIRC	0.032	[0.026, 0.038]	-0.095	47,316	11,829/35,487	$< 10^{-12}$	Yes
INFE	CONG	0.001	[-0.000, 0.003]	-0.000	57,396	14,349/43,047	0.092	No
INFE	DIGE	0.046	[0.038, 0.053]	-0.043	50,496	12,624/37,872	$< 10^{-12}$	Yes
INFE	ENDO	-0.007	[-0.012, -0.001]	-0.041	52,912	13,228/39,684	0.019	Yes
INFE	GEN	0.034	[0.027, 0.041]	-0.019	54,228	13,557/40,671	$< 10^{-12}$	Yes
INFE	INJ	0.018	[0.014, 0.023]	-0.003	56,776	14,194/42,582	$< 10^{-12}$	Yes
INFE	MENT	0.019	[0.014, 0.024]	-0.015	55,708	13,927/41,781	$< 10^{-12}$	Yes
INFE	MUSC	0.067	[0.059, 0.074]	-0.034	53,160	13,290/39,870	$< 10^{-12}$	Yes
INFE	NEOP	0.012	[0.008, 0.016]	-0.005	56,392	14,098/42,294	2.37×10^{-8}	Yes
INFE	NERV	0.041	[0.035, 0.047]	-0.013	55,164	13,791/41,373	$< 10^{-12}$	Yes
INFE	PERI	-0.000	[-0.001, 0.001]	-0.000	57,440	14,360/43,080	0.840	No
INFE	RESP	0.070	[0.061, 0.078]	-0.028	53,088	13,272/39,816	$< 10^{-12}$	Yes
INFE	SKIN	0.023	[0.018, 0.027]	-0.003	56,176	14,044/42,132	$< 10^{-12}$	Yes
INFE	SUPP	0.044	[0.036, 0.052]	-0.036	51,696	12,924/38,772	$< 10^{-12}$	Yes
INJ	BLD	0.012	[0.008, 0.015]	-0.002	67,792	16,948/50,844	1.38×10^{-12}	Yes
INJ	CIRC	-0.003	[-0.008, 0.002]	-0.084	55,432	13,858/41,574	0.259	No
INJ	CONG	0.000	[-0.001, 0.002]	-0.000	68,164	17,041/51,123	0.545	No
INJ	DIGE	0.085	[0.078, 0.092]	-0.036	61,068	15,267/45,801	$< 10^{-12}$	Yes
INJ	ENDO	0.016	[0.012, 0.021]	-0.042	62,584	15,646/46,938	8.61×10^{-12}	Yes
INJ	GEN	0.037	[0.031, 0.043]	-0.019	65,540	16,385/49,155	$< 10^{-12}$	Yes
INJ	INFE	0.014	[0.010, 0.017]	-0.003	67,788	16,947/50,841	$< 10^{-12}$	Yes
INJ	MENT	0.007	[0.003, 0.011]	-0.012	66,008	16,502/49,506	5.19×10^{-4}	Yes
INJ	MUSC	0.025	[0.018, 0.032]	-0.022	62,736	15,684/47,052	2.60×10^{-12}	Yes
INJ	NEOP	0.012	[0.008, 0.015]	-0.003	67,260	16,815/50,445	7.92×10^{-11}	Yes
INJ	NERV	0.049	[0.044, 0.055]	-0.013	65,736	16,434/49,302	$< 10^{-12}$	Yes
INJ	PERI	0.000	[-0.000, 0.000]	0.000	68,220	17,055/51,165	0.094	No
INJ	RESP	0.042	[0.034, 0.050]	-0.031	64,212	16,053/48,159	$< 10^{-12}$	Yes
INJ	SKIN	0.014	[0.010, 0.019]	-0.004	67,400	16,850/50,550	6.71×10^{-10}	Yes
INJ	SUPP	0.053	[0.046, 0.060]	-0.028	62,260	15,565/46,695	$< 10^{-12}$	Yes
MENT	BLD	0.013	[0.009, 0.018]	-0.004	41,676	10,419/31,257	6.12×10^{-9}	Yes
MENT	CIRC	0.004	[-0.004, 0.012]	-0.131	32,900	8,225/24,675	0.397	No

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Table 3 – continued from previous page

Exposure	Outcome	ATE	95% CI	Placebo	n	n_T/n_C	q-value	FDR
MENT	CONG	0.001	[-0.001, 0.003]	-0.001	41,992	10,498/31,494	0.380	No
MENT	DIGE	0.084	[0.075, 0.093]	-0.038	36,348	9,087/27,261	$< 10^{-12}$	Yes
MENT	ENDO	-0.007	[-0.014, -0.000]	-0.042	38,216	9,554/28,662	0.048	Yes
MENT	GEN	0.048	[0.040, 0.056]	-0.022	40,132	10,033/30,099	$< 10^{-12}$	Yes
MENT	INFE	0.011	[0.006, 0.016]	-0.004	41,672	10,418/31,254	1.11×10^{-5}	Yes
MENT	INJ	0.019	[0.014, 0.024]	-0.004	41,444	10,361/31,083	1.04×10^{-11}	Yes
MENT	MUSC	0.100	[0.091, 0.109]	-0.039	38,508	9,627/28,881	$< 10^{-12}$	Yes
MENT	NEOP	0.011	[0.007, 0.016]	-0.005	41,408	10,352/31,056	1.81×10^{-6}	Yes
MENT	NERV	0.076	[0.069, 0.083]	-0.018	40,192	10,048/30,144	$< 10^{-12}$	Yes
MENT	PERI	0.001	[0.000, 0.001]	-0.000	42,036	10,509/31,527	4.79×10^{-3}	Yes
MENT	RESP	0.048	[0.039, 0.058]	-0.034	39,340	9,835/29,505	$< 10^{-12}$	Yes
MENT	SKIN	0.006	[0.001, 0.012]	-0.008	41,668	10,417/31,251	0.031	Yes
MENT	SUPP	0.075	[0.066, 0.085]	-0.037	37,128	9,282/27,846	$< 10^{-12}$	Yes
MUSC	BLD	0.012	[0.010, 0.014]	-0.003	135,756	33,939/101,817	$< 10^{-12}$	Yes
MUSC	CIRC	0.008	[0.004, 0.012]	-0.123	120,917	30,230/90,687	2.03×10^{-4}	Yes
MUSC	CONG	0.000	[-0.001, 0.001]	-0.000	136,160	34,040/102,120	0.688	No
MUSC	DIGE	0.067	[0.062, 0.072]	-0.032	128,880	32,220/96,660	$< 10^{-12}$	Yes
MUSC	ENDO	0.026	[0.023, 0.030]	-0.035	130,256	32,564/97,692	$< 10^{-12}$	Yes
MUSC	GEN	0.032	[0.027, 0.037]	-0.022	133,472	33,368/100,104	$< 10^{-12}$	Yes
MUSC	INFE	0.010	[0.007, 0.013]	-0.002	135,864	33,966/101,898	$< 10^{-12}$	Yes
MUSC	INJ	0.019	[0.017, 0.022]	-0.002	135,152	33,788/101,364	$< 10^{-12}$	Yes
MUSC	MENT	0.010	[0.008, 0.013]	-0.014	134,052	33,513/100,539	$< 10^{-12}$	Yes
MUSC	NEOP	0.001	[-0.001, 0.004]	-0.006	135,364	33,841/101,523	0.330	No
MUSC	NERV	0.014	[0.011, 0.018]	-0.014	133,720	33,430/100,290	$< 10^{-12}$	Yes
MUSC	PERI	-0.000	[-0.000, 0.000]	-0.000	136,228	34,057/102,171	0.212	No
MUSC	RESP	0.039	[0.034, 0.045]	-0.028	132,304	33,076/99,228	$< 10^{-12}$	Yes
MUSC	SKIN	0.015	[0.011, 0.018]	-0.006	135,620	33,905/101,715	$< 10^{-12}$	Yes
MUSC	SUPP	0.042	[0.036, 0.048]	-0.033	129,768	32,442/97,326	$< 10^{-12}$	Yes
NEOP	BLD	0.037	[0.032, 0.042]	-0.004	33,712	8,428/25,284	$< 10^{-12}$	Yes
NEOP	CIRC	0.006	[-0.003, 0.015]	-0.099	26,968	6,742/20,226	0.204	No
NEOP	CONG	0.002	[0.000, 0.004]	-0.000	34,152	8,538/25,614	0.022	Yes
NEOP	DIGE	0.087	[0.076, 0.098]	-0.041	29,556	7,389/22,167	$< 10^{-12}$	Yes
NEOP	ENDO	0.038	[0.030, 0.046]	-0.040	31,004	7,751/23,253	$< 10^{-12}$	Yes
NEOP	GEN	0.101	[0.091, 0.110]	-0.012	31,464	7,866/23,598	$< 10^{-12}$	Yes
NEOP	INFE	0.025	[0.020, 0.031]	-0.002	33,876	8,469/25,407	$< 10^{-12}$	Yes
NEOP	INJ	0.029	[0.023, 0.035]	-0.003	33,692	8,423/25,269	$< 10^{-12}$	Yes
NEOP	MENT	0.011	[0.005, 0.017]	-0.016	33,032	8,258/24,774	1.18×10^{-3}	Yes
NEOP	MUSC	0.073	[0.063, 0.084]	-0.042	31,536	7,884/23,652	$< 10^{-12}$	Yes
NEOP	NERV	0.027	[0.019, 0.036]	-0.018	32,844	8,211/24,633	2.96×10^{-10}	Yes
NEOP	PERI	0.000	[-0.000, 0.001]	-0.000	34,192	8,548/25,644	0.172	No
NEOP	RESP	0.050	[0.039, 0.061]	-0.028	31,880	7,970/23,910	$< 10^{-12}$	Yes
NEOP	SKIN	0.030	[0.024, 0.036]	-0.002	33,552	8,388/25,164	$< 10^{-12}$	Yes
NEOP	SUPP	0.091	[0.080, 0.102]	-0.036	29,832	7,458/22,374	$< 10^{-12}$	Yes
NERV	BLD	0.003	[-0.000, 0.006]	-0.004	86,074	21,519/64,555	0.094	No
NERV	CIRC	-0.021	[-0.026, -0.016]	-0.109	70,998	17,750/53,248	$< 10^{-12}$	Yes
NERV	CONG	0.001	[0.000, 0.003]	-0.000	86,610	21,653/64,957	0.014	Yes
NERV	DIGE	0.066	[0.060, 0.073]	-0.052	78,398	19,600/58,798	$< 10^{-12}$	Yes
NERV	ENDO	0.035	[0.031, 0.040]	-0.044	79,510	19,878/59,632	$< 10^{-12}$	Yes
NERV	GEN	0.067	[0.061, 0.072]	-0.019	83,798	20,950/62,848	$< 10^{-12}$	Yes
NERV	INFE	0.008	[0.005, 0.012]	-0.004	86,186	21,547/64,639	3.48×10^{-6}	Yes
NERV	INJ	0.022	[0.018, 0.027]	-0.007	85,942	21,486/64,456	$< 10^{-12}$	Yes
NERV	MENT	0.014	[0.010, 0.017]	-0.016	83,966	20,992/62,974	$< 10^{-12}$	Yes
NERV	MUSC	0.042	[0.036, 0.048]	-0.043	81,314	20,329/60,985	$< 10^{-12}$	Yes
NERV	NEOP	0.006	[0.003, 0.009]	-0.005	85,794	21,449/64,345	3.22×10^{-4}	Yes
NERV	PERI	0.000	[0.000, 0.000]	0.000	86,670	21,668/65,002	1.28×10^{-3}	Yes
NERV	RESP	0.059	[0.049, 0.070]	-0.037	81,838	20,460/61,378	$< 10^{-12}$	Yes
NERV	SKIN	0.027	[0.023, 0.031]	-0.004	86,090	21,523/64,567	$< 10^{-12}$	Yes
NERV	SUPP	0.077	[0.070, 0.084]	-0.037	80,038	20,010/60,028	$< 10^{-12}$	Yes

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Table 3 – continued from previous page

Exposure	**Outcome**	**ATE**	**95% CI**	**Placebo**	n	n_T/n_C	q**-value**	**FDR**
RESP	BLD	0.004	[0.002, 0.006]	-0.003	160,806	40,202/120,604	1.07×10^{-3}	Yes
RESP	CIRC	0.053	[0.050, 0.057]	-0.109	144,466	36,117/108,349	$< 10^{-12}$	Yes
RESP	CONG	-0.001	[-0.002, -0.000]	-0.000	161,198	40,300/120,898	0.058	No
RESP	DIGE	0.038	[0.033, 0.043]	-0.039	153,418	38,355/115,063	$< 10^{-12}$	Yes
RESP	ENDO	0.008	[0.005, 0.012]	-0.045	154,718	38,680/116,038	3.13×10^{-6}	Yes
RESP	GEN	-0.007	[-0.011, -0.003]	-0.021	158,402	39,601/118,801	1.86×10^{-3}	Yes
RESP	INFE	0.010	[0.007, 0.012]	-0.003	160,838	40,210/120,628	4.15×10^{-11}	Yes
RESP	INJ	0.006	[0.003, 0.009]	-0.004	160,586	40,147/120,439	1.38×10^{-4}	Yes
RESP	MENT	0.012	[0.009, 0.014]	-0.018	158,650	39,663/118,987	$< 10^{-12}$	Yes
RESP	MUSC	0.017	[0.013, 0.022]	-0.039	157,470	39,368/118,102	$< 10^{-12}$	Yes
RESP	NEOP	0.000	[-0.002, 0.002]	-0.004	160,394	40,099/120,295	0.787	No
RESP	NERV	0.040	[0.036, 0.044]	-0.015	158,634	39,659/118,975	$< 10^{-12}$	Yes
RESP	PERI	-0.000	[-0.000, 0.000]	-0.000	161,230	40,308/120,922	0.841	No
RESP	SKIN	0.005	[0.003, 0.008]	-0.004	160,550	40,138/120,412	4.23×10^{-4}	Yes
RESP	SUPP	0.048	[0.041, 0.054]	-0.043	155,022	38,756/116,266	$< 10^{-12}$	Yes
SKIN	BLD	0.003	[-0.000, 0.007]	-0.004	62,812	15,703/47,109	0.068	No
SKIN	CIRC	0.016	[0.010, 0.022]	-0.095	52,028	13,007/39,021	1.98×10^{-7}	Yes
SKIN	CONG	0.002	[0.001, 0.004]	-0.000	63,200	15,800/47,400	4.01×10^{-4}	Yes
SKIN	DIGE	0.021	[0.013, 0.028]	-0.044	57,008	14,252/42,756	3.99×10^{-8}	Yes
SKIN	ENDO	0.013	[0.008, 0.018]	-0.038	58,344	14,586/43,758	2.17×10^{-7}	Yes
SKIN	GEN	0.016	[0.010, 0.023]	-0.024	60,608	15,152/45,456	1.65×10^{-6}	Yes
SKIN	INFE	0.018	[0.014, 0.022]	0.001	62,672	15,668/47,004	$< 10^{-12}$	Yes
SKIN	INJ	0.026	[0.021, 0.030]	-0.003	62,472	15,618/46,854	$< 10^{-12}$	Yes
SKIN	MENT	0.004	[0.000, 0.009]	-0.014	61,256	15,314/45,942	0.053	No
SKIN	MUSC	0.055	[0.048, 0.063]	-0.042	59,412	14,853/44,559	$< 10^{-12}$	Yes
SKIN	NEOP	0.014	[0.010, 0.018]	-0.004	62,372	15,593/46,779	$< 10^{-12}$	Yes
SKIN	NERV	0.043	[0.036, 0.049]	-0.016	61,104	15,276/45,828	$< 10^{-12}$	Yes
SKIN	PERI	0.000	[0.000, 0.001]	0.000	63,220	15,805/47,415	6.65×10^{-4}	Yes
SKIN	RESP	0.050	[0.040, 0.060]	-0.030	59,556	14,889/44,667	$< 10^{-12}$	Yes
SKIN	SUPP	0.070	[0.063, 0.077]	-0.036	58,356	14,589/43,767	$< 10^{-12}$	Yes
SUPP	BLD	0.011	[0.009, 0.013]	-0.001	161,332	40,333/120,999	$< 10^{-12}$	Yes
SUPP	CIRC	0.010	[0.007, 0.014]	-0.087	144,596	36,149/108,447	6.72×10^{-9}	Yes
SUPP	CONG	0.001	[0.000, 0.001]	-0.000	161,736	40,434/121,302	0.051	No
SUPP	DIGE	0.039	[0.034, 0.045]	-0.033	153,340	38,335/115,005	$< 10^{-12}$	Yes
SUPP	ENDO	0.019	[0.016, 0.022]	-0.042	155,064	38,766/116,298	$< 10^{-12}$	Yes
SUPP	GEN	0.017	[0.012, 0.021]	-0.016	158,796	39,699/119,097	$< 10^{-12}$	Yes
SUPP	INFE	0.007	[0.004, 0.009]	-0.003	161,380	40,345/121,035	8.98×10^{-6}	Yes
SUPP	INJ	0.021	[0.019, 0.024]	-0.002	160,772	40,193/120,579	$< 10^{-12}$	Yes
SUPP	MENT	0.006	[0.004, 0.009]	-0.016	159,108	39,777/119,331	1.19×10^{-6}	Yes
SUPP	MUSC	0.032	[0.027, 0.036]	-0.028	157,196	39,299/117,897	$< 10^{-12}$	Yes
SUPP	NEOP	-0.004	[-0.006, -0.002]	-0.003	160,828	40,207/120,621	4.59×10^{-4}	Yes
SUPP	NERV	0.028	[0.023, 0.032]	-0.017	158,952	39,738/119,214	$< 10^{-12}$	Yes
SUPP	PERI	-0.000	[-0.000, 0.000]	-0.000	161,828	40,457/121,371	0.871	No
SUPP	RESP	0.014	[0.007, 0.020]	-0.028	156,620	39,155/117,465	8.29×10^{-5}	Yes
SUPP	SKIN	0.002	[-0.001, 0.005]	-0.005	161,068	40,267/120,801	0.261	No

S5. Additional Robustness Diagnostics

This section reports practical diagnostics used to assess the empirical support and stability of the edge-level ATE estimates.

Overlap and effective sample size. Edges with median propensity scores away from the extremes and with larger effective sample sizes under inverse-probability weighting are generally more stable candidates for interpretation, as they indicate better empirical overlap between treated and control observations. These quantities do not prove identification, but they are important practical diagnostics for distinguishing well-supported edges from statistically more fragile ones.

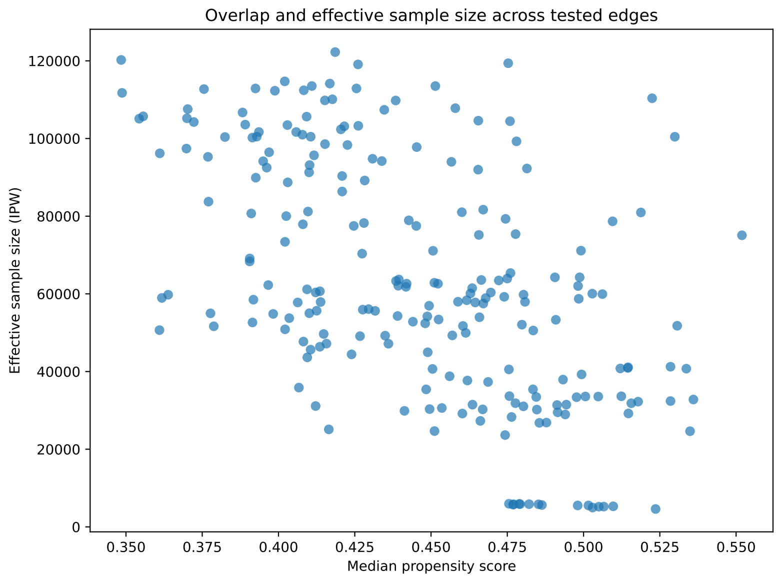

Fig. 2. Overlap and effective sample size diagnostics across tested edges. Median propensity score and effective sample size under inverse-probability weighting are shown for the evaluated $D_a \to D_b$ relations.

Placebo diagnostics. Placebo effects are computed in a short post-index, pre-follow-up window and are used as a residual-bias diagnostic. Large placebo activity may indicate that part of the estimated association still reflects surveillance intensity, coding synchronization, or care-process structure rather than a delayed downstream progression signal. We therefore use placebo as an interpretive filter rather than as a strict exclusion rule.

S6. Implementation and Hyperparameter Configuration

This section summarizes the main implementation choices used in the current `CHRONOS` pipeline, including both model hyperparameters and operational trial-design settings. We report them together because, in the present study, the final edge estimates depend jointly on representation-learning choices, optimization settings, and the protocol used to instantiate each edge-specific target trial from the longitudinal records. Table 4 is organized into five blocks: network architecture, optimization settings, causal loss weights, contrastive-learning parameters,

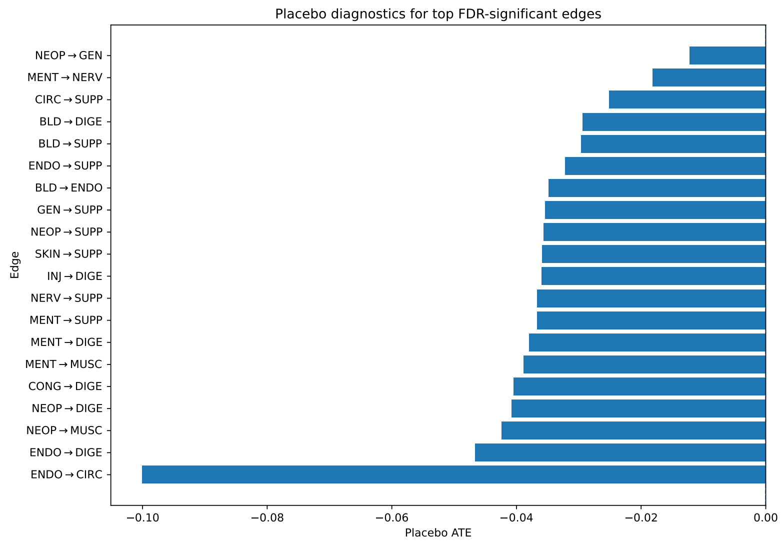

Fig. 3. Placebo diagnostics for representative FDR-significant edges. Edges with larger placebo activity are interpreted more cautiously in the main text.

and edge-level trial design choices. The *Network Architecture* block describes the main representational capacity of the model. In particular, `CHRONOS` uses a latent embedding dimension of 64 and a single-layer GRU encoder, which provide a compact sequential representation while keeping the architecture relatively lightweight. A dropout rate of 0.1 is used for regularization, GELU is adopted as the activation function, and the mutual-information critics use a hidden dimension of 128. The *Optimization* block reports the core training configuration. The model is trained with AdamW using learning rate 1×10^{-3}, batch size 1024, and 10 epochs. Cross-fitting is performed with $K = 5$ folds so that treatment-effect estimation is always evaluated out of fold. In the implemented pipeline, propensity scores are clipped at $\epsilon = 0.01$ before AIPW aggregation to reduce instability due to extreme inverse-probability weights. The *Causal Loss Weights* and *Contrastive Params* blocks specify the relative contribution and operationalization of the auxiliary representation-learning objectives. The propensity-related loss is weighted by $\lambda_{prop} = 1.0$, and the effect-guided contrastive regularization by $\alpha = 0.5$. For the mutual-information objective, we follow the notation of the main text and write $\mathcal{L}_{\mathrm{MI}} = \lambda_b \widehat{MI}(\mathbf{Z}, T) - \lambda_o \widehat{MI}(\mathbf{Z}, Y)$, while in the implementation we denote these two coefficients by β and γ, respectively. Accordingly, $\beta = 0.1$ controls treatment-information suppression and $\gamma = 0.01$ controls outcome-information preservation. The contrastive margin is set to $m = 1.0$, similarity is defined through the 30th percentile threshold of the batchwise effect-distance distribution, and at most 2048 pairs are constructed per batch. Finally, the *Trial Design* block reports the operational cohort-construction rules used for the edge-level target trial emulation procedure. Each edge-specific trial uses a 180-day baseline window, a 180-day treatment washout, a 30-day outcome washout, a 14-day post-index buffer, a 365-day follow-up horizon, a 60-day active-observation requirement, 1:3 risk-set control sampling, and a minimum treated support threshold of 50. These values correspond to the actual defaults used in the current training and trial-launch scripts. The mutual-information component is used as an auxiliary regularizer and is applied intermittently rather than at every optimization step: the `MINE` critics are updated in separate auxiliary steps and the resulting penalty is injected periodically into the main objective. Second, although `CHRONOS` is conceptually described as a Siamese framework, the current implementation realizes this behavior through a shared single-patient encoder combined with batchwise pair construction in latent space, rather than through an explicit dual-branch forward pass.

Table 4. Implementation settings, hyperparameters, and operational trial-design choices used in `CHRONOS`.

Category	**Parameter**	**Value**
Network Architecture	Latent Dimension (z)	64
	RNN Layers (GRU)	1
	Dropout Rate	0.1
	Activation Function	GELU
	MI Critic Hidden Dim	128
Optimization	Learning Rate (η)	1×10^{-3}
	Optimizer	AdamW
	Batch Size	1024
	Training Epochs	10
	Cross-fitting Folds (K)	5
	Propensity Clipping (ϵ)	0.01
Causal Loss Weights	Propensity Weight (λ_{prop})	1.0
	Contrastive Weight (α)	0.5
	MI Balancing Weight ($\beta \equiv \lambda_b$)	0.1
	MI Preservation Weight ($\gamma \equiv \lambda_o$)	0.01
Contrastive Params	Contrastive Margin (m)	1.0
	Similarity Percentile (p)	30%
	Max Pairs per Batch	2048
Trial Design	Baseline Window (W_a)	180 days
	Treatment Washout	180 days
	Outcome Washout (W_B)	30 days
	Buffer Period (Δ)	14 days
	Follow-up Horizon (h)	365 days
	Active Observation Requirement	60 days
	Control Ratio (k)	1:3
	Min. Treated per Edge	50

S7. Sensitivity Analysis of Trial-Design Choices

In addition to architectural ablations, we assessed the robustness of the estimated progression effects with respect to key design choices in the edge-level target-trial emulation procedure. In particular, we varied three design parameters: *(i)* the post-index buffer period ($\Delta = 14$ vs. 30 days), *(ii)* the follow-up horizon used to define the outcome window ($h = 365$ vs. 730 days), and *(iii)* the treated-to-control sampling ratio used to construct the matched risk sets (1 : 1 vs. 1 : 3).

These sensitivity analyses were intended to evaluate whether the main findings depend strongly on the exact temporal specification of the trial design or on the amount of control sampling used to support the treated–control comparison.

Table 5 provides the aggregate entry point for interpreting the results. Each row corresponds to one alternative trial-design setting, and the columns should be read jointly rather than in isolation. *Edges* reports the number of representative relations evaluated in the sensitivity subset, while *FDR Yes* indicates how many remain significant after global multiple-testing correction. *Mean ATE* summarizes the average estimated effect across the evaluated edges, *Placebo* reports the median absolute placebo magnitude as a compact diagnostic of short-window residual bias, and *ESS* reports the median effective sample size as an indicator of the stability of the weighted comparison. Accordingly, a more favorable specification is one that preserves significant edges while maintaining interpretable effect magnitudes, limited placebo activity, and adequate effective sample support.

Table 6 complements this aggregate view with edge-level estimates. Each row corresponds to one directed progression relation and each column to one design setting. Within each cell, the first value is the estimated ATE and the value in parentheses is the corresponding placebo estimate; boldface indicates that the edge remains significant after FDR correction. Since the evaluated subset is small, this table is mainly useful for understanding *which* relations are sensitive to changes in design choices, rather than for drawing conclusions from any single edge in isolation.

Table 5. Sensitivity analysis summary across the evaluated trial-design configurations. *FDR Yes* denotes the number of edges retained after global FDR correction, *Placebo* reports the median absolute placebo estimate, and *ESS* reports the median effective sample size.

Setting	Edges	FDR Yes	Mean ATE	Placebo	ESS
Buffer 14d	3	1	0.053	0.019	386.1
Buffer 30d	3	1	0.056	0.025	386.1
Follow-up 365d	3	1	0.053	0.019	386.1
Follow-up 730d	3	2	0.022	0.019	386.1
Control ratio 1:1	3	0	-0.006	0.000	193.3
Control ratio 1:3	3	1	0.053	0.019	386.1

Table 6. Edge-level sensitivity results. Each cell reports *ATE (Placebo)* for the corresponding design setting. Boldface indicates that the edge remains significant after global FDR correction.

Edge	Buf 14d	Buf 30d	$h = 365$d	$h = 730$d	Ratio 1:1	Ratio 1:3
`BLD` → `ENDO`	**0.133** (0.000)	**0.133** (0.000)	**0.133** (0.000)	**0.133** (0.000)	0.005 (0.000)	**0.133** (0.000)
`MENT` → `MUSC`	0.052 (-0.021)	0.064 (-0.042)	0.052 (-0.021)	0.018 (-0.021)	0.018 (0.000)	0.052 (-0.021)
`NERV` → `GEN`	-0.025 (-0.019)	-0.030 (-0.025)	-0.025 (-0.019)	**-0.086** (-0.019)	-0.041 (-0.003)	-0.025 (-0.019)

Several patterns emerge from these results. First, varying the buffer period from 14 to 30 days has only a limited impact on the overall results. The aggregate summaries remain very similar, with the same number of FDR-retained edges, nearly identical effective sample size, and only modest differences in mean ATE and placebo magnitude. At the edge level, the main positive signal `BLD` → `ENDO` is unchanged, while the other two relations show only moderate numerical variation. This suggests that the main findings are not strongly driven by the exact choice of post-index exclusion window within this range.

Second, extending the follow-up horizon from 365 to 730 days does not materially affect ESS or placebo diagnostics, but it does alter the pattern of retained effects. In particular, the number of FDR-significant edges increases from 1 to 2, yet the mean ATE across the evaluated subset decreases from 0.053 to 0.022. This occurs because the longer horizon introduces a more heterogeneous mix of edge-level behaviors: while `BLD` → `ENDO` remains unchanged, `MENT` → `MUSC` attenuates, and `NERV` → `GEN` becomes significantly negative. For this reason, the 730-day specification should not be interpreted as simply "better" because it retains more significant edges; rather, it changes the composition and direction of the detected signals.

Third, the treated-to-control sampling ratio has the clearest impact on empirical support. Moving from a $1:3$ to a $1:1$ ratio approximately halves the median ESS (from 386.1 to 193.3) and removes all FDR-significant findings in this subset. At the edge level, this change is especially visible for `BLD` → `ENDO`, whose estimated effect drops from a stable significant value of 0.133 under $1:3$ sampling to a near-null nonsignificant value of 0.005 under $1:1$. This pattern is consistent with the expected role of control sampling in edge-level causal discovery, where adequate control support is important for maintaining stable overlap and sufficient power.

Taken together, these sensitivity analyses indicate that the main `CHRONOS` findings are reasonably robust to moderate changes in temporal design parameters, especially for the strongest positive relation in the representative subset. At the same time, they highlight that control sampling is a more consequential design choice for statistical stability than modest adjustments to the buffer period or follow-up horizon. More broadly, the results suggest that trial-design sensitivity in this setting concerns not only whether an edge remains significant, but also how effect magnitude and direction evolve under alternative specifications.

S8. Epidemiological Baseline via Edge-Specific Cox Models

As an additional epidemiological reference, we complemented the main `CHRONOS` analysis with edge-specific Cox proportional hazards models. The purpose of this analysis was not to replace the proposed causal framework, but to provide a conventional time-to-event baseline that may be more familiar to clinical and epidemiological readers.

A Cox proportional hazards model is a standard survival-analysis method used to study whether an exposure is associated with a different rate of experiencing an event over time. In our setting, for each directed trajectory $D_a \to D_b$, the incident occurrence of disease category D_a defines the exposure, and the event is the first observed onset of the downstream disease category D_b after the edge-specific index date. The Cox model asks whether treated patients tend to experience the downstream event sooner than control patients during follow-up. Its main summary measure is the *hazard ratio* (HR), which compares the event rate in treated versus control patients over time. An HR greater than 1 indicates that the event tends to occur more quickly or more frequently in the treated group, an HR below 1 indicates a lower event rate in the treated group, and an HR close to 1 indicates little difference in event timing between the two groups.

This estimand differs from the one used in the main `CHRONOS` analysis. Whereas the primary framework targets confounding-adjusted risk differences under edge-level target-trial emulation, the Cox model summarizes association through relative event timing. For this reason, the Cox baseline should be interpreted as a complementary epidemiological layer rather than as a direct replacement for the main causal estimand.

To preserve comparability with the main analysis, each Cox model was fitted on the same emulated cohort used for the corresponding directed trajectory $D_a \to D_b$. Therefore, the Cox baseline inherited the same trial-design settings adopted in the main `CHRONOS` pipeline: a 180-day baseline window, a 180-day treatment washout, a 30-day outcome washout, a 14-day post-index buffer, a 365-day follow-up horizon, a 60-day active-observation requirement, 1:3 risk-set control sampling, and a minimum treated support threshold of 50. In this way, the Cox analysis remained aligned with the same operational cohort-construction rules used for the target-trial emulation procedure, differing mainly in the statistical modeling layer and in the effect measure being reported.

For each candidate trajectory $D_a \to D_b$, the exposure variable corresponded to the trial-specific treatment indicator, and time-to-event was defined as the elapsed time from the edge-specific index date to the first occurrence of the outcome category D_b within follow-up. Patients without an observed outcome onset during the horizon were administratively censored at the end of follow-up. Hazard ratios, 95% confidence intervals, and associated p-values were estimated for the exposure effect, and false discovery rate correction was applied across evaluated trajectories using the Benjamini–Hochberg procedure.

When interpreting Cox results, both the HR and its confidence interval are important. An HR substantially above 1 suggests that the downstream event occurs earlier or more often in treated patients, whereas an HR substantially below 1 suggests the opposite. However, very large or very small HR values can also arise from sparse events, near-separation, or numerical instability. For this reason, Cox estimates should be interpreted more cautiously when confidence intervals are wide, when outcome events are rare, or when the fitted model is close to singularity.

Because the Cox models were fitted independently across a large number of edge-specific emulated cohorts, numerical stability varied substantially across trajectories. In particular, some edge-specific cohorts exhibited sparse outcome counts, no events in one arm, limited variability in baseline covariates, near-separation, or singular design matrices. As a result, not all Cox fits were identifiable or numerically stable. We therefore interpret the Cox analysis as a complementary epidemiological reference rather than as the primary inferential layer of the study.

Overall, Cox models were attempted for all $15 \times 14 = 225$ directed $D_a \to D_b$ trajectories. Valid estimates were obtained for a subset of edges, whereas the remaining trajectories were skipped or flagged because of singular matrices, no outcome events during follow-up, non-convergence, or non-finite hazard-ratio estimates. This behavior is expected in large-scale edge-wise survival analyses, especially when the same emulated design is reused across many heterogeneous trajectories with widely varying event support. Importantly, statistical significance in the Cox analysis should not be over-interpreted in isolation. Some apparently extreme HR estimates were driven by very small numbers of events or quasi-separation and are therefore not clinically meaningful despite nominal or even FDR-adjusted significance. For this reason, interpretation should focus primarily on trajectories with adequate event support, non-degenerate treated and control event rates, finite and non-extreme confidence intervals, and overall directional consistency with the main confounding-adjusted `CHRONOS` estimates.

Table 7. Summary of the edge-specific Cox baseline analysis across all $D_a \to D_b$ trajectories.

Quantity	**Value**
Total $D_a \to D_b$ trajectories evaluated	225
Valid Cox fits	168
Skipped or non-identifiable fits	57
FDR-significant valid fits	10
Stable interpretable fits after support filtering	134
Stable and FDR-significant fits	4

Table 7 provides an aggregate overview of the Cox baseline. The total number of attempted trajectories reflects the full directed screening space, while the remaining rows summarize how many models yielded numerically usable estimates and how many remained statistically supported after multiple-testing correction. In practice, this table should be read mainly as a diagnostic of feasibility and stability: it shows that Cox modeling can be applied successfully to a substantial subset of edges, but is considerably more fragile than the main doubly robust framework when deployed at scale over many heterogeneous trajectories.

Table 8 reports a small subset of the most stable and interpretable Cox results. Each row corresponds to one directed trajectory. The *HR* column reports the estimated hazard ratio for the treated versus control comparison; values above 1 indicate a higher downstream event rate in treated patients, and values below 1 indicate a lower event rate. The *95% CI* column summarizes estimation uncertainty, the *q-value* reports FDR-adjusted significance, and *Events* gives the total number of observed downstream outcome events supporting the estimate. In practice, the most interpretable rows are those with adequate event support and finite, non-extreme confidence intervals.

Table 8. Representative stable Cox baseline results for selected $D_a \to D_b$ trajectories.

Edge	**HR**	**95% CI**	**q-value**	**Events**
`INJ` → `MUSC`	2.51	[1.60, 3.96]	0.0016	100
`SKIN` → `DIGE`	0.468	[0.304, 0.720]	0.0118	181
`INJ` → `DIGE`	0.507	[0.328, 0.784]	0.0378	145
`RESP` → `GEN`	0.718	[0.581, 0.888]	0.0378	565

The representative results in Table 8 illustrate both the usefulness and the limitations of the Cox baseline. On the one hand, these trajectories yield finite and reasonably supported hazard-ratio estimates, showing that at least a subset of the detected progression relations also corresponds to measurable differences in event timing under a standard survival-model formulation. On the other hand, the relatively small number of stable and FDR-significant edges confirms that large-scale edge-wise survival modeling is substantially more fragile than the main doubly robust trial-emulation framework.

Taken together, these findings suggest that the Cox baseline is most useful as a secondary epidemiological reference for selected trajectories rather than as a universal screening tool across all directed disease-pair trials. Its role in this study is therefore to complement the interpretation of the primary `CHRONOS` estimates, not to replace them.

To provide a more direct comparison between the proposed causal framework and the epidemiological baseline, we also considered directional agreement between the edge-specific `CHRONOS` ATE estimates and the Cox hazard ratios. Because the two analyses target different quantities—absolute risk differences under doubly robust target-trial emulation in the main framework versus hazard ratios under a survival-model formulation in the baseline—they should not be compared numerically on the same scale. Instead, the relevant question is whether they support a compatible *direction* of the progression signal.

In practice, directional concordance was assessed by checking whether edges with positive confounding-adjusted ATE estimates under `CHRONOS` also tended to show hazard ratios above 1 under the Cox baseline, and conversely whether edges with weaker or opposite Cox signals were associated with smaller or less compelling progression effects in the main analysis. This comparison is intended as a qualitative consistency check rather than as a formal equivalence analysis.

Table 9 reports a small subset of representative trajectories for which both the `CHRONOS` estimate and the Cox baseline were available in stable form. Each row corresponds to one directed exposure–outcome trajectory. The *ATE* column reports the confounding-adjusted absolute risk difference estimated by `CHRONOS`, whereas the *HR* column reports the hazard ratio from the edge-specific Cox model. Because these two quantities measure different aspects of the longitudinal association, they should not be compared numerically on the same scale. Instead, the table is intended to support a qualitative comparison of *direction*: positive ATEs are more naturally aligned with HR values above 1, whereas negative ATEs are more naturally aligned with HR values below 1. The final column, *Concordance*, indicates whether the two methods support a compatible orientation of the progression signal under this criterion. Accordingly, the table should be read as an interpretability-oriented consistency check rather than as a formal agreement analysis.

Table 9. Comparison between representative `CHRONOS` ATE estimates and Cox hazard ratios.

Exposure	Outcome	ATE	HR	95% CI	Concordance
`INJ`	`MUSC`	0.025	2.51	[1.60, 3.96]	Yes
`SKIN`	`DIGE`	0.021	0.468	[0.304, 0.720]	No
`INJ`	`DIGE`	0.085	0.507	[0.328, 0.784]	No
`RESP`	`GEN`	-0.007	0.718	[0.581, 0.888]	Yes

Table 9 highlights that agreement between `CHRONOS` and the Cox baseline is only partial, which is consistent with the fact that the two approaches summarize different aspects of the same trajectory. For `INJ` $\rightarrow$ `MUSC`, both methods support the same qualitative direction: the positive ATE under `CHRONOS` is matched by an HR substantially above 1, suggesting that this trajectory is associated both with higher downstream risk and with faster event occurrence in the treated group. A similar directional agreement is observed for `RESP` $\rightarrow$ `GEN`, where the slightly negative ATE is accompanied by an HR below 1.

By contrast, `SKIN` $\rightarrow$ `DIGE` and `INJ` $\rightarrow$ `DIGE` show directional discordance: the `CHRONOS` ATE is positive, whereas the Cox HR is below 1. These discrepancies do not necessarily imply that one of the two analyses is wrong. Rather, they indicate that the two methods are sensitive to different features of the longitudinal signal. `CHRONOS` targets an absolute confounding-adjusted risk difference under the emulated trial design, whereas the Cox model summarizes relative hazard over follow-up and is more directly influenced by event timing, censoring structure, and survival-model instability.

For this reason, the table should be read mainly as a qualitative consistency check rather than as a head-to-head validation exercise. Agreement in direction increases confidence that a trajectory is robust across two distinct modeling views, while disagreement indicates that the corresponding edge may be more sensitive to modeling assumptions, sparse-event structure, or the distinction between absolute risk and relative timing. In this sense, the Cox baseline serves as an interpretability-oriented epidemiological reference, not as a competing causal estimator.

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